Hyperlocal Courier Dispatch & Tracking Engine

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Requirements Table

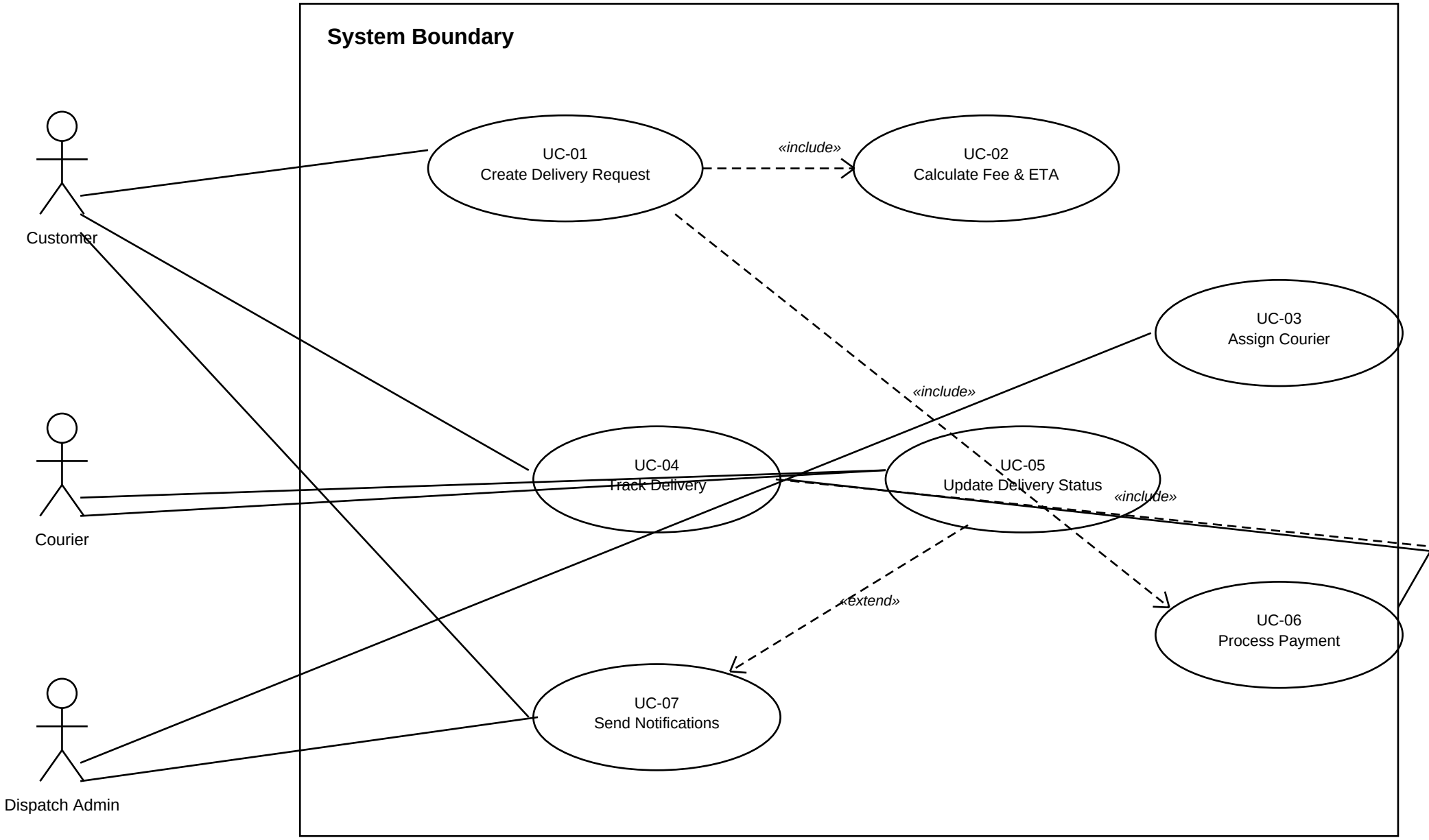
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a customer to create a courier delivery request by entering pickup location, drop location, package details, and contact information.	High	A request is created only when all mandatory fields are valid; a unique request ID is displayed.	Core function for initiating a courier delivery.	Verify mandatory fields and unique request ID.
FR-002	Functional	The system shall calculate the estimated delivery fee and delivery time based on pickup/drop locations, package details, and available courier options.	High	For a valid request, the system displays a fee and ETA before dispatch confirmation.	Customers need cost and ETA before confirming.	Define pricing and ETA inputs clearly.
FR-003	Functional	The system shall assign an available courier to a confirmed delivery request and notify the courier of the pickup and drop details.	High	When an eligible courier is available, the request changes to Assigned and the courier receives the delivery details.	Required to dispatch deliveries in the hyperlocal area.	Assignment must prevent duplicate allocation.
FR-004	Functional	The system shall allow customers to track the current status and latest location of an active delivery.	High	For an active request, the customer can view status and a latest-location update; completed requests show Delivered.	Provides visibility during delivery.	Location freshness should be measurable.
FR-005	Functional	The system shall allow a courier to update delivery status through Accepted, Picked Up, In Transit, and Delivered.	High	Each valid status update is stored with a timestamp and the next permitted status is shown.	Creates an auditable delivery lifecycle.	Reject invalid status transitions.
NFR-001	Nonfunctional	The system shall display the dispatch/booking confirmation response within 3 seconds for at least 95% of valid requests under normal load.	High	A performance test over 100 valid requests records at least 95 responses in ≤ 3 seconds.	Fast response is important for real-time dispatch.	Use server-side timing logs.
NFR-002	Nonfunctional	The system shall protect customer and courier data using authenticated access and encrypted communication.	High	Unauthenticated users cannot access protected delivery records, and all application traffic uses HTTPS/TLS.	Delivery records contain contact and location information.	Verify access-control and TLS tests.

Actors and Use Cases

- Actors: Customer, Courier, Dispatch Admin, Payment Gateway, Map/Location Service.
- Use Cases: UC-01 Create Delivery Request; UC-02 Calculate Fee & ETA; UC-03 Assign Courier; UC-04 Track Delivery; UC-05 Update Delivery Status; UC-06 Process Payment; UC-07 Send Notifications.
- Relationships: Create Delivery Request «include» Calculate Fee & ETA; Create Delivery Request «include» Process Payment; Track Delivery «include» Retrieve Latest Location; Update Delivery Status «extend» Report Delivery Issue.

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UML Use-Case Diagram



Legend: Solid line = association Dashed arrow = «include» / «extend»

Use-Case Flow – Assign & Dispatch Courier

Use Case ID: UC-03
Primary Actor: Dispatch Admin
Supporting Actors: Courier, Map/Location Service, Notification Service

Preconditions

- A valid delivery request has been created and booking authorization is available.
- The dispatch system can access current courier availability.

Main Success Scenario

1. Dispatch Admin opens the list of confirmed delivery requests.

2. System displays request details, pickup/drop locations, package details, ETA, and eligible nearby couriers.

3. Dispatch Admin selects an eligible courier.

4. System verifies that the courier is available and not already assigned to another active request.

5. System assigns the courier and changes the delivery status to Assigned.

6. System sends pickup and drop details to the courier.

7. Courier accepts the assignment.

8. System records the assignment timestamp and displays the updated status to the customer.

9. Use case ends successfully.

Alternate Flow – No Courier Available

3a. No eligible courier is available within the configured service area.

3a1. System marks the request as Awaiting Courier and displays the current estimated wait time.

3a2. System notifies the customer that dispatch is delayed.

3a3. When a suitable courier becomes available, the request returns to the assignment step.

Postcondition: The request is assigned to a courier or remains in Awaiting Courier with a recorded status and notification.